UC DAVIS VETERINARY MEDICINE

Hedgehogs and Sugar Gliders

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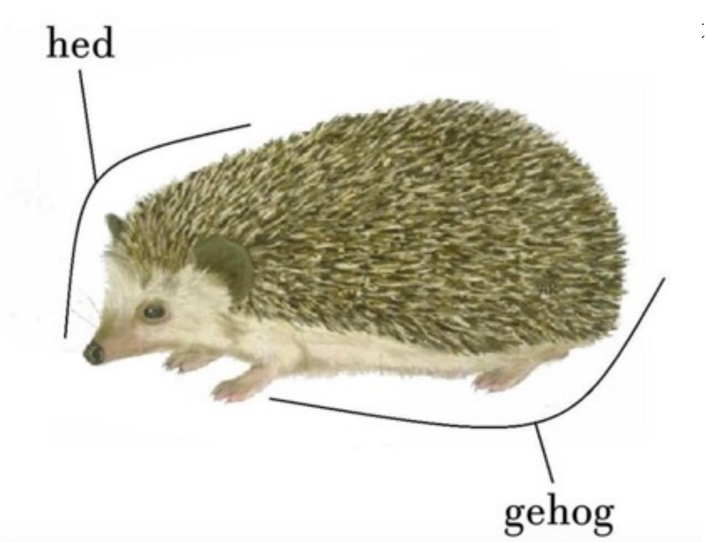
INTRODUCTION

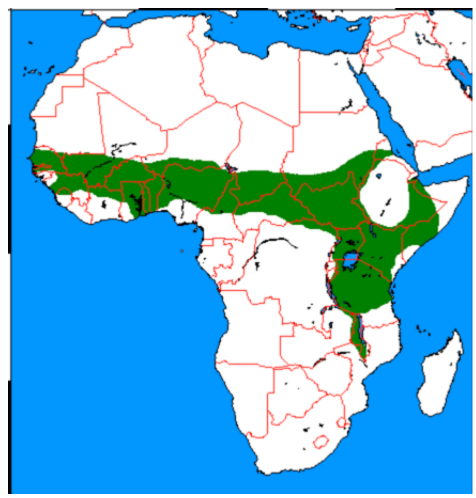
Welcome to the wonderful world of hedgehogs and sugar gliders! During our VET436 course, we hope to be able to share the most important clinical aspects to be aware of when managing these types of animals. This includes knowing about their unique anatomy, general husbandry, and most common diseases one may encounter in a clinical setting. Should you wish to learn more about these species, the "African Pygmy Hedgehogs" and "Sugar Gliders" chapters in Quesenberry's [Ferrets, Rabbits, and Rodents: Clinical Medicine and Surgery] can help further expand your knowledge base as well.

HEDGEHOG BIOLOGY AND SPECIAL ANATOMY

We will begin by discussing hedgehog biology and anatomy.



The two most commonly-encountered hedgehog species are the central African hedgehog (*Atelerix albiventris*) and the European hedgehog (*Erinaceus europaeus*). The central African hedgehog, also known as the "African pygmy" hedgehog, is native to the savannah and steppe regions of central and eastern Africa.

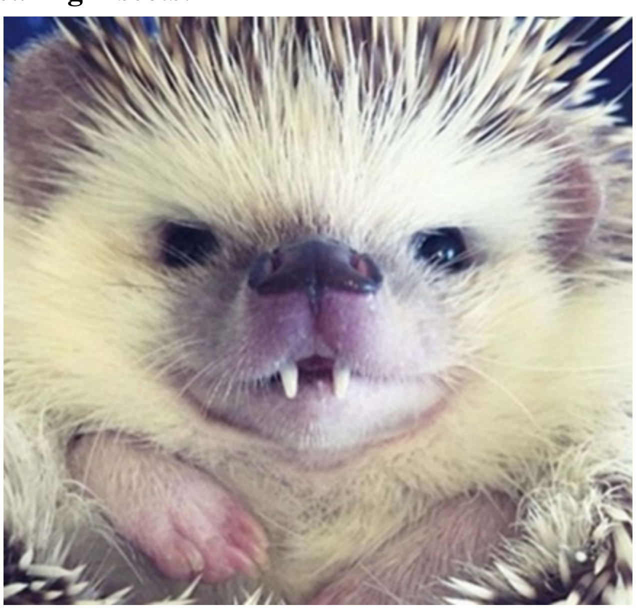


In the pet trade, Atelerix species are generally referred to as African pygmy hedgehogs. Unfortunately due to human movement of these animals, they have become invasive species in other countries such as New Zealand and have caused damage to native species such as insects, snails, lizards, and ground-nesting birds. Because of this, **there is a ban on importation of these animals, and pet hedgehogs in the United States are captive-bred**. Even within the United States, hedgehogs are illegal to own in states like California for fear of damage to native wildlife if they were to escape into the wild, and veterinarians and owners alike are encouraged to check state USDA regulations regarding the legality of owning hedgehogs in a given state.

Knowing about hedgehog biology is helpful when it comes to providing optimal husbandry and care in either a household or clinical setting. **Hedgehogs are nocturnal animals, and generally should be housed alone due to their territorial nature when fully-grown.**

They are an **omnivorous species, but primarily insectivorous while consuming plant matter opportunistically.** Hedgehogs do not hibernate, but **can enter a state of dormancy called torpor when they feel their environmental conditions are not of appropriate temperature, either too cold or excessively hot.** With torpor, hedgehogs become significantly less reactive to external stimuli, have decreased respiratory and heart rates, and can be markedly ataxic if prompted to move while in this state. Torpor can last for several weeks and predisposes an animal to increased susceptibility to infection, so it is important to properly educate owners to help prevent their hedgehogs from entering torpor due to poor environmental conditions. **In captivity, hedgehogs typically live 4-6 years, but can live up to 8 years if well cared for.**

Beginning from the snout and working our way back, there are several unique features of hedgehog dentition that are worth noting. Hedgehogs have **anelodont (limited growth period), brachydont (closed-rooted) teeth.** The first incisor in each quadrant is large and projects forward. **The mandibular first incisors occlude into a space between the maxillary first incisors, which is well-suited in their arrangement to spearing insects.**



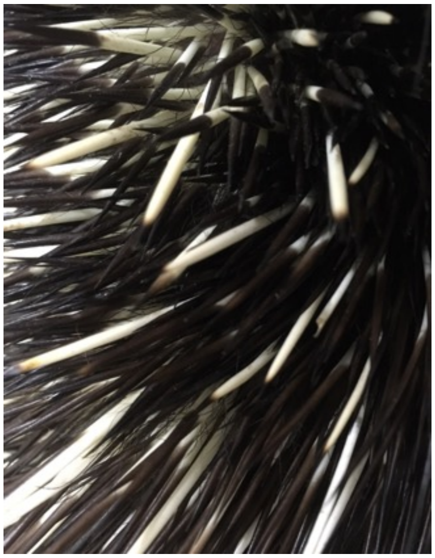
This noticeable gap between the first maxillary incisor teeth is significantly larger in female than in male African pygmy hedgehogs, which may be because of the female's overall larger size and prehension of relatively larger prey items. The molar teeth are relatively flattened with multiple cusps and are suited for crushing food items.

The stomach is simple, and a vomiting reflex is present. There is no cecum, and the colon is smooth. Feces are normally soft but formed.

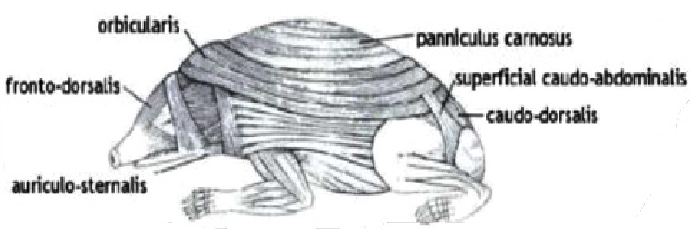
The crown and dorsum of hedgehogs (which is collectively called the mantle) are covered in thousands of smooth spines.

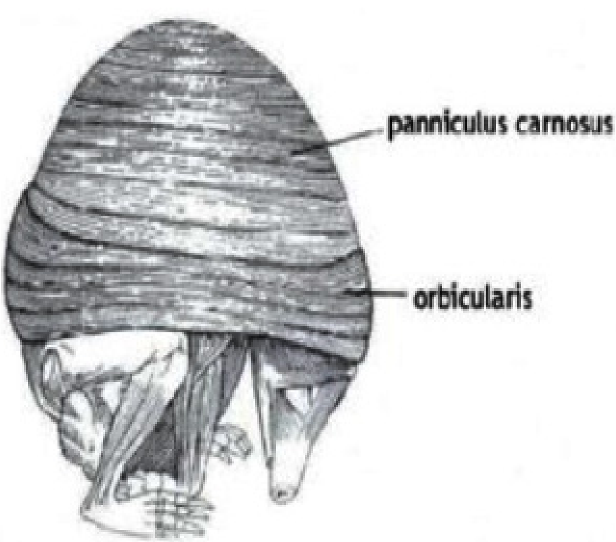
Hedgehog spines are composed of keratin and have a complex internal structure that confers lightness, strength, and elasticity. Each spine has a round basal bulb that firmly attaches it within the follicle, whereas a more narrowed portion at the skin surface allows each spine to bend when force is applied. Healthy spines are difficult to pull from the follicle without breaking at this narrowed portion. Spines are absent from the midline of the crown, and hair on the feet and muzzle is sparse to absent. **These spines should NOT be con-**

fused with the quills in large rodent species such as porcupines, and should not readily detach from the body like porcupine quills do when in contact with another animal or surface.



Instead, a hedgehog will contract a series of muscles and "ball up" if frightened by a noise or touch. **The major muscles of the mantle are the fronto-dorsalis and caudo-dorsalis, and when these muscles contract, they pull the head down and the rump up. The panniculus carnosus muscle then pulls and wraps around the spine, and the orbicularis muscle lastly closes like a pouch.**





A hedgehog may remain rolled up for hours with relatively little muscular effort, which can make examination very challenging!

Hedgehogs have a very sensitive olfactory system and well-developed olfactory lobes. The vomeronasal organ is also prominent, and hedgehogs may occasionally exhibit a flehmen response. It is believed that olfaction plays a key role in a wild hedgehog's ability to navigate within its home range, detect food, avoid predators, and communicate with other hedgehogs. Hedgehogs also have sensitive hearing, though their vision is not as well-developed. Foraging hedgehogs normally emit a variety of snuffling sounds. Frightened or agitated hedgehogs make a distinctive hissing sound that may be punctuated with various puffing and cough-like sounds. **Hedgehogs also demonstrate a unique behavior called self-anointing or anting.**

This behavior may be elicited by a variety of substances, particularly those with a strong or unusual odor. During this behavior, a hedgehog will take the material or object into its mouth, mix it with frothy saliva, and apply the mixture to its spines with its tongue. Though the exact function of this behavior is unknown, it is hypothesized that this helps hedgehogs to blend in with their environmental surroundings or spread strong or seemingly toxic scents along their spines to ward off predators when curled.

HUSBANDRY

If socialized from a young age, hedgehogs can make interactive and rewarding pets, but husbandry practices should always be reviewed with owners. As previously-mentioned, hedgehogs are nocturnal, so they become more active in the evenings and overnight. They should be housed alone and should be provided with plenty of space to allow for exercise and placement of enrichment items such as running wheels, tunnels, and hiding spaces. Healthy hedgehogs are very active, and as large a cage as possible should be provided (2 × 3 feet are minimal floor dimensions). Hedgehogs are able to climb and escape through small holes, so the cage must be secure and lidded as well. In the wild, these animals normally roam miles and miles in search of food and water, so they enjoy having a spacious area in captivity too! **Heat support is also important to avoid torpor, especially in areas that become cold in the fall and winter seasons; ideal temperature range for them is 75-85 degrees Fahrenheit.** Their enclosures should be well-ventilated to prevent concentration of aerosols from waste materials, and easily-cleaned substrates such as compressed pellets or fleece are recommended. Clay, sand, or other clumping litters may become ingested or stuck to the underside of a hedgehog, and so should NOT be used.

In terms of their diet, **hedgehogs should ideally be on a commercial hedgehog food that is appropriately balanced for their species, such as those produced by Mazuri or Exotic Nutrition**.

These diets should be higher in protein and lower in fat; low-fat cat foods can also be introduced if needed. Hedgehogs require a **higher level of dietary fiber than carnivores**, which may be related to the large quantity of insect exoskeletons consumed in their natural diet in the wild. Live insects should be supplemented into the diet whenever possible, particularly for enrichment, and should focus on more protein-rich insect species such as mealworms and crickets. Other insects such as waxworms should be more sparingly given due to their higher fat content. Fresh water should be provided at all times.

PREVENTATIVE MEDICINE AND PHYSICAL EXAMINATION

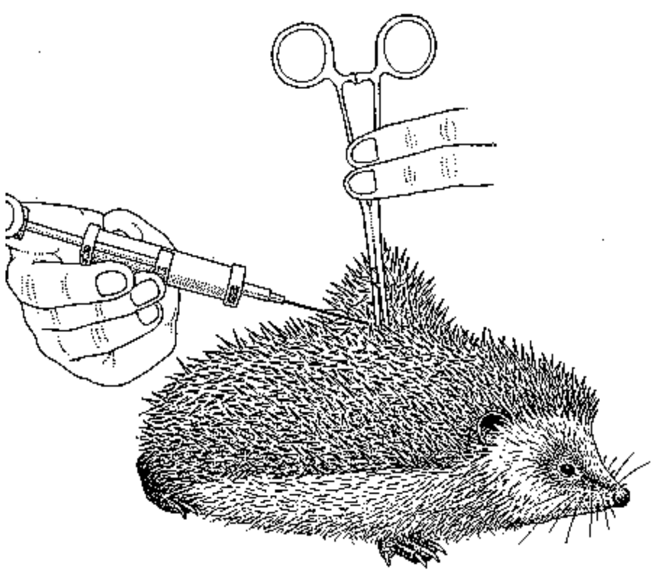
And here's where we veterinarians really come into play! Annual exams for any species are opportunities to make sure an animal is healthy and that husbandry is appropriate, especially for our exotic companion animals. **Care recommendations should include a thorough physical examination, including an oral examination, as this species is unfortunately very prone to neoplasia.** Many veterinarians will actually recommend twice-yearly examinations in order to screen for neoplasia. Dental prophylaxis is recommended for older hedgehogs, and tartar control and hard food treats/diets may help. No vaccinations are required, but annual complete blood count and chemistry panel are recommended to identify for any early signs of disease. Based upon presentation and history, radiographs and fecal analysis/culture can be considered as well.

Given their natural defense mechanisms, hedgehogs can be difficult to fully examine while awake. One can do a cursory visual exam in a clear plastic container and friendly hedgehogs may even allow auscultation and abdominal palpation. Once uncurled, hedgehogs may be restrained by holding the spined skin as one would hold the scruff of other species. To do this, one can grasp the hedgehog just behind the ears at the junction of the haired skin and mantle. Scruffing can provide enough access to perform auscultation, abdominal palpation, or cephalic blood collection.

However, scruffing is not commonly done and can cause excess stress. Thus, **use of brief sedation or anesthesia is often most helpful to allow for complete**

examination and sample collection.

If needing to administer treatments injectably, injection sites in hedgehogs are actually very similar to available sites in other mammalian species with a few hedgehog-specific guidelines. In general, **all fluids should be warmed prior to administration, as use of cold fluids can negatively decrease core body temperature.** Subcutaneous injections can be performed into either the spined or furred areas. **Spined areas (along the dorsum) have low vascularity, and although easily accessible during handling of the non-anesthetized hedgehog, fluids and medications administered at this location may not be absorbed for hours. Furred areas (along the lateral body wall or ventrum) have high vascularity and fluids and medications administered here will be rapidly absorbed, however, this site is not typically available in stressed animals.**



Subcutaneous injections can also be given at the junction of the furred and spined skin. Intramuscular injections can be performed into the epaxial muscles, the larger muscles of the cranial/caudal thigh, or into the orbicularis muscles. Abundant adipose tissue may lead to delayed uptake or distribution if overlying muscular areas. Both intravenous and intraosseous injections in hedgehogs require deep sedation or anesthesia and are reserved for intensive cases such as shock resuscitation, surgical intervention, or therapy for seizing animals.

Phlebotomy, IV/IO catheterization, and injection sites for African pygmy hedgehogs[10,14,17]

Reported Phlebotomy Sites[a]	IV Catheterization Sites[a]	IO Catheterization Sites[a]	Injection Sites
Cranial vena cava Jugular vein Femoral vein Lateral saphenous Cephalic	Cephalic	Proximal tibia, through tibial crest Proximal femur, through trochanteric fossa	IM: triceps, quadriceps, gluteal SQ: flank, junction of furred and spined skin midbody

Venipuncture from proximal jugular vein: See also video.

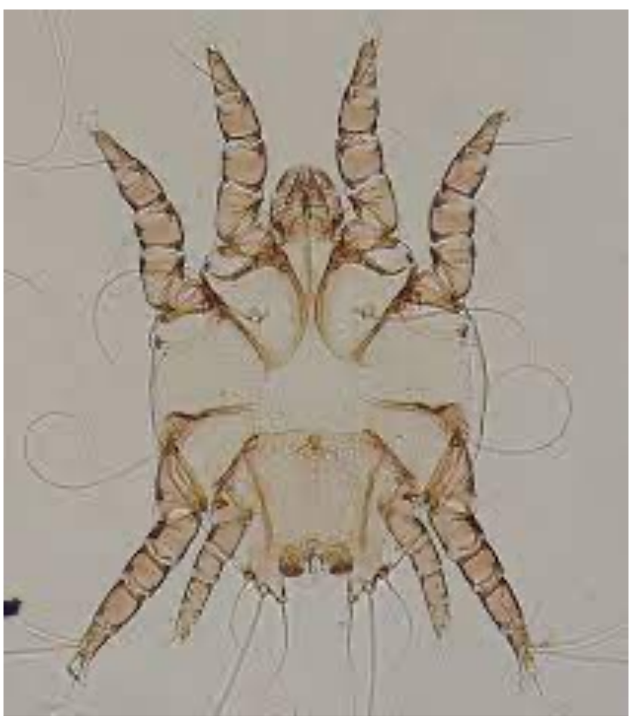
ANALGESIA

There is some research being done evaluating the antinociceptive effects of different opioids for pain management in hedgehogs. Methadone seems to provide short-term analgesia, approximately 2 hours with some transient behavioral changes. While hydromorphone provided longer periods of analgesia, approximately 6 hours, it also had transient abnormal behaviors such as vocalization, chewing motions of the jaw, and paw raising and it also decreased the total food intake. Buprenorphine provided long-lasting antinociception, approximately 48 hours, without apparent adverse effects.

COMMON DISEASES

Dermatitis

Hedgehogs can present for a variety of dermatologic issues, particularly dermatitis. **In pet hedgehogs, the most common cause of dermatitis is acariasis**. ***Caparinia tripilis* has been identified more from captive-bred hedgehogs, whereas *C. erinacei* is more likely to cause disease in wild and wild-caught African pygmy hedgehogs.**



Other possible mites include *Chorioptes* spp. and *Notoedres* spp., although Caparinia spp. are more common. Signs of dermatitis can be severe and include pruritus, spine loss or damage, hyperkeratitis, seborrhea, and white or brown crusts around the spines from mite droppings.

Additionally, specific areas such as the ears may be affected. Lethargy, decreased appetite, and self-mutilation can also be common sequelae. Differentials for spine loss and skin abnormalities includes dermatophytosis (typically from *Trichophyton erinacei, T. Mentagrophytes*, and *Microsporum* spp.), allergies including contact dermatitis, neoplasia (papilloma, SCC, LSA, sebaceous gland carcinoma, osteosarcoma), and yeast dermatitis. Other causes of pruritus include in-growing spines, which can be appreciated in young hedgehogs. Hedgehogs can also develop pemphigus foliaceus and pododermatitis.

All work-ups of an animal presenting for a dermatological complaint should receive a physical examination, CBC, biochemistry panel, and discussion of husbandry. Sometimes evidence of immunosuppression or disease in other organ systems can be highlighted and may be underlying causes of disease. Additional derm-specific diagnostics include a skin scrape (can be diagnostic for mites), DTM culture (dermatophytosis), and spine/skin cytology and culture to assess for bacterial infection. Therapies for dermatitis should focus on the etiology of the disease. In cases of acariasis, several antiparasitics have been employed including ivermectin, fipronil spray, and selamectin. **Most importantly, environmental decontamination must be performed**

to ensure that the mites in the environment are eliminated. In cases of dermatophytosis, topical products containing chlorhexidine, terbinafine, or some other antifungal agent can be used. However, many times systemic antifungal therapy is necessary for a cure. Itraconazole is the current treatment of choice for dermatophytosis in small exotic mammals and terbinafine can also be considered as a safe systemic therapy. **All animals in a household may need to be treated, even if not showing signs of disease.** Anti-inflammatories such as meloxicam and antihistamines may be implemented to improve patient comfort, and antibiotics may be needed to combat any secondary bacterial infection as well.

Diarrhea

Diarrhea can also be a common presenting complaint for pet hedgehogs. It is a non-specific sign and there can be a wide range of etiologies.

Infectious causes include:

- **Salmonellosis: Approximately 28% of hedgehogs are non-clinical but have Salmonella cultured from feces. In some, the presence of this microbe may cause clinical signs like weight loss, decreased appetite, dehydration, lethargy, and death. This is an important zoonotic pathogen and can cause serious disease in humans.**
- Alimentary candidiasis has been reported in a hedgehog that presented with diarrhea, weight loss, and blood in the stool.
- Fatal cryptosporidiosis has been reported to cause lesions in the small and large intestines in juvenile hedgehogs.

Non-infectious causes include:

- Obstructions from inappropriate ingestion of items such as rubber, hair, or carpet fibers. Animals with obstructions may or may not vomit. A single case of mesenteric torsion has been described.
- Inappropriate diet or recent dietary change. Affected animals may have a suggestive history of being fed inappropriate dietary items, especially milk.
- Neoplasia of the gastrointestinal tract has been reported, particularly lymphosarcoma.
- A variety of other underlying systemic diseases can lead to diarrhea, including hepatic or renal diseases.

Diagnostics for work-up for diarrhea should include a thorough husbandry review to focus on inappropriate dietary items or a recent change in diet, minimum database (physical exam, bloodwork, +/- radiographs), and fecal testing such as culture, cytology, and/or flotation based on your clinical suspicions. In cases of suspect salmonellosis, a fecal culture should be performed. **It is important to indicate to the laboratory that you are interested in identification of Salmonella, as this can be best performed with the utilization of special selective media. However, remember that healthy hedgehogs without diarrhea can be Salmonella positive on culture and should not be treated if not clinical, as this potentiates antibiotic resistance! Client education is imperative in cases of Salmonellosis or another zoonotic diseases. Clients must understand the risk to young and immunocompromised children and older individuals and diligently practice the basics of hand-washing and sanitation.**

Fecal cytology can help to characterize a monopopulation of bacteria (which is abnormal) or fungal organisms as is the case in candidiasis, and can be done in-house for little cost. If cryptosporidium is suspected, acid-fast staining (or PCR) should be pursued. Fecal flotation should be pursued to rule out parasites, although most parasites afflict wild animals or new purchases.

In treating a hedgehog with diarrhea, it is important to treat the underlying cause if one has been identified. Potential treatments may involve use of antibiotics (bacterial infections), antifungals (for issues such as Candidiasis), or may be more surgical in cases of intestinal obstruction or neoplasia. Underlying systemic diseases such as hepatic or renal disease should also be treated based upon origin. Broad spectrum treatment regimes for hepatic disease are similar to that for dogs and cats and include nutritional support, fluid support, antibiosis, and hepatoprotectants. Treatment of renal disease focuses primarily on fluid support and, when indicated, antibiotic therapy.

Neoplasia

Neoplasia is exceedingly common in pet African pygmy hedgehogs, with tumor types in every body system described. This higher incidence is due largely to the genetic bottleneck previously experiences by the species. **In retrospective reports, 29-50% of post-mortem submissions of hedgehogs exhibit neoplasia, with integumentary and hemolymphatic lesions being the most common.** Unfortunately the majority of these tumors are malignant (85%) and carry a poor prognosis. **The three most common tumors in pet hedgehogs are oral squamous cell carcinoma, lymphosarcoma, and mammary gland tumors** (adenocarcionoma; most likely in females > 3 years old).

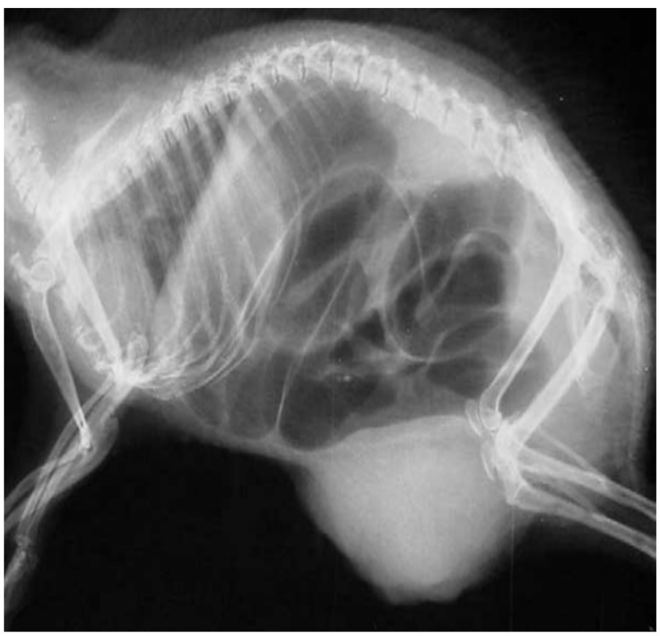
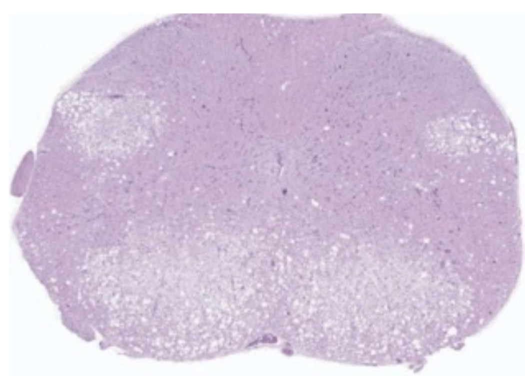


Figure 1. Radiographic view of a mammary adenocarcinoma in an adult female African hedgehog (*Atelerix albiventris*).

Neurologic Disease

An ataxic hedgehog is another common reason for presentation to a veterinarian. Signs described by the owner may include difficulty walking and poor control over the limbs. **The three major causes of an ataxic or otherwise neurologically inappropriate hedgehog include Wobbly Hedgehog Syndrome, intervertebral disc disease, and spinal neoplasia. Another very important cause to always keep on the radar is rabies, which has been reported in one hedgehog.** Other less-common causes may include vascular compromise, toxin exposure, trauma, otitis media/interna, hypocalcemia, hepatic/renal encephalopathy, infectious causes (Baylisascaris), or torpor in the face of excessively cold conditions.

The pathogenesis of Wobbly Hedgehog Syndrome involves a demyelinating paralysis that often begins caudally and gradually ascends cranially. The disease occurs in approximately 10% of hedgehogs in North America, most commonly younger animals <2 years of age. Clinical signs usually begin with owner reports of mild progressive ataxia (stumbling, wobbling, tripping) that become more progressive over a periods of weeks to months. As the disease progresses, signs such as tremors, exophthalmos, scoliosis, seizures, muscle atrophy, and self-mutilation can also be appreciated. Paralysis ascends from hindlimbs to forelimbs in about 70% of cases, and ultimately leads to complete paralysis. Dysphagia occurs in the terminal stages of the disease, and **death occurs within 18-25 months after onset of initial clinical signs. Unfortunately for affected hedgehogs, there is no cure for Wobbly Hedgehog Syndrome,** and treatment is focused on supportive care (hand-feeding, fluid therapy, treatment of secondary infections). Euthanasia is recommended when hedgehogs become immobilized, and **diagnosis is confirmed on post-mortem examination only. Usually there are no gross lesions, but histopathology reveals a spongiform myelinopathy and vacuolization of the white matter of the cerebrum, cerebellum, brainstem, and spinal cord. Inflammation of the CNS is NOT seen.**

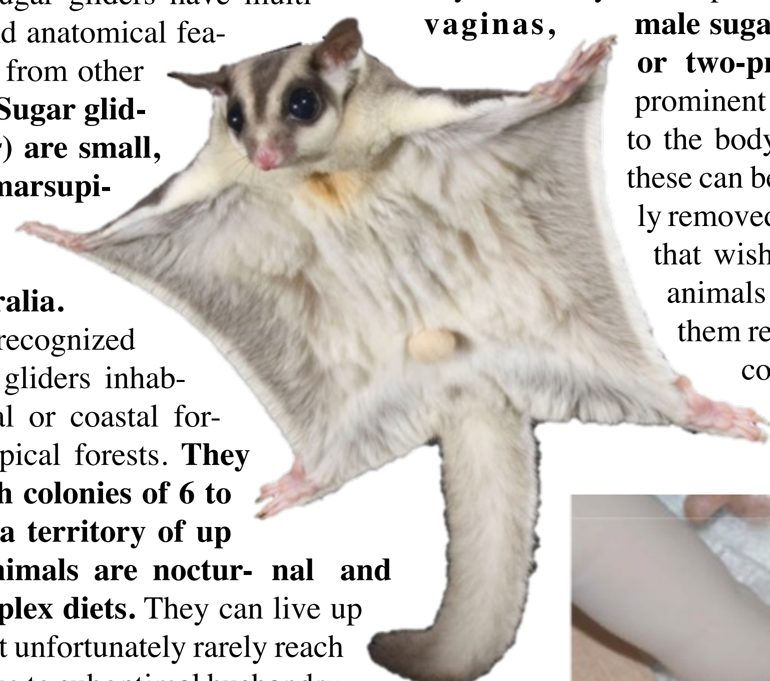


Intervertebral disc disease in hedgehogs occurs in animals >4 years old, with males and females both being affected. Clinical signs include chronic progressive ataxia, lameness, proprioceptive deficits, and bladder atony. Multiple discs are often affected, and can sometimes be seen as narrowed disc spaces on radiographs. For animals that are assessed post-mortem, necropsy often reveals degeneration of the nucleus pulposus and annulus fibrosus, as well as disc mineralization. **Cervical lesions may be more common than thoracolumbar.** Temporary treatment with corticosteroids has been described, but is primarily focused on maintaining patient comfort levels.

Due to the high prevalence of neurological disorders, a neurological exam can be performed. Normal neurological exam in hedgehogs has been described. Most aspects of the standard neurologic examination can be performed with some limitations. Overall, evaluation of the pelvic lumbs can be hard to performed. Hedgehogs lack menace response but they display contraction of the frontodorsalis musculе instead.

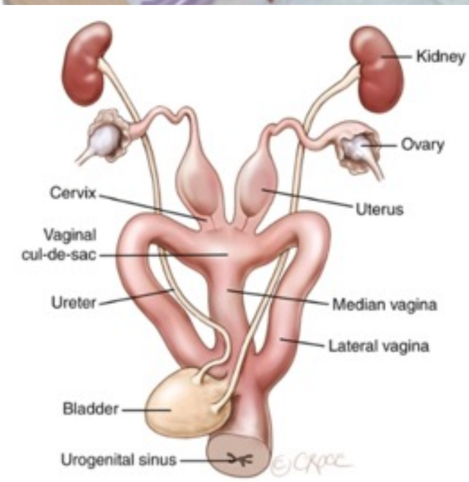
SUGAR GLIDER BIOLOGY AND SPECIAL ANATOMY

Just like hedgehogs, sugar gliders have multiple unique biological and anatomical features that separate them from other small mammal species. **Sugar gliders (*Petaurus breviceps*) are small, nocturnal arboreal marsupials that are native to New Guinea and the eastern coast of Australia.** There are at least seven recognized subspecies. Wild sugar gliders inhabit open areas in tropical or coastal forests and dry, inland tropical forests. **They are social animals, with colonies of 6 to 10 animals occupying a territory of up to 2.5 acres. These animals are noctur- nal and have omnivorous, complex diets.** They can live up to 10-12 years of age, but unfortunately rarely reach those ages in captivity due to suboptimal husbandry. Also similar to hedgehogs, sugar gliders are technically illegal to own in California, but veterinarians are allowed to treat them if presented in a clinical setting.



Moving onto anatomy, **sugar gliders are marsupials, and so they are most well known for the presence of a pouch.** The pouch is designed for raising young. **Female sugar gliders have a well-developed, mid-abdominal pouch (marsupium) which contains four teats capable of secreting different compositions of milk depending on the joey's age.** The adrenal glands of females are twice the size of male adrenals based on body weight. Sugar glider females generally raise 1-2 young at a time. **The pouch is also clinically important to examine for the presence of joeys or mammary masses** (but most will not allow examination while awake, so use of anesthesia will often be helpful).

The reproductive anatomy of sugar gliders is very intriguing. **Females actually have two lateral vaginas, 2 cervices, and 2 uteri, with the vaginas opening into a single "cul-de-sac" at their proximal portions.** There is also a median vagina through which a joey travels once it is developmentally progressed enough to journey externally to the pouch. **To adapt to the lateral vaginas, male sugar gliders also have a bifid, or two-pronged, penis.** They have prominent testicles that are kept close to the body during consciousness, but these can be readily and easily surgically removed under anesthesia for clients that wish to house male and female animals together without the risk of them reproducing. The cloaca is the common terminal opening of the rectum, urinary ducts, and genital ducts.



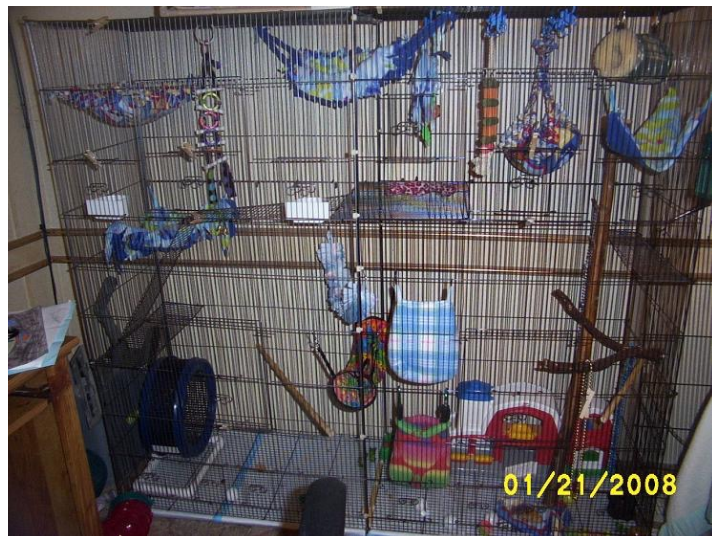
The dentition of sugar gliders is comprised of brachydont (closed-rooted) teeth with short crowns and well-developed roots. Their incisors are specialized for chewing, while their premolar and molar teeth are more for crushing food materials. They have a well-developed cecum to allow for microbial fermentation of complex polysaccharides, and do not appear as sensitive to the potential dysbiotic effects of antibiotic therapy compared to other small mammal species.

The anatomical aspect that really puts the "glide" in sugar gliders is their patagium, which is a thin membrane that extends from the wrist to the ankle. This structure remains folded during ambulation, but can then be spread wide via extension of all 4 limbs to allow sugar gliders to glide short distances. The patagium should be inspected for tears during physical examination. Their front feet each have five digits, all with claws, and there is a lengthened fourth digit on each for extracting insects from crevices. There are also five digits on each rear foot, all containing a sharp claw except the first digit (hallux), which is opposable. Digits two and three on the hind foot are identical in length and partially fused (syndactylous) to form a grooming comb.

Sugar gliders also have multiple prominent scent glands. These scent glands are involved in group recognition and communication among gliders. Males have frontal and sternal scent glands. Females have scent glands in their pouch. Both have paracloacal glands, which produce a white oily secretion with a musky odor resembling soured fruit. Dominant males will have androgen-sensitive frontal, gular, and paracloacal scent glands, while females have paracloacal and pouch scent glands. These will have a greasy secretion, especially during breeding season, and should not be confused with dermatitis or neoplastic lesions.

HUSBANDRY

In regards to husbandry, **sugar gliders should not be housed alone, and should be kept in pairs at minimum. Sugar gliders are another small mammal species that can enter a torpor state if kept too cold, and so ideal environmental temperatures should fall between 65-90 degrees Fahrenheit.** Sugar gliders are very active at night, and should be housed in multi-level enclosures with plenty of environment items to encourage exercise and natural behaviors such as climbing and jumping. They also enjoy wearing down their incisors on wood, paper products, and chew ropes. Substrate can be inclusive of recycled paper products such as Carefresh, fleece, or regular shredded paper.



In the wild, sugar gliders have a complex diet that varies with the seasons. Saps, gums, nectar, and pollen, as well as various insects and arachnids, are typically ingested. **Sugar gliders are omnivores rather than insectivores, and so should be offered a varied diet in captivity.** Glider teeth are designed to compress, not shear, insects and food materials. Gliders extract the nutrients within hemolymph and soft tissues of arthropods, and discard the less digestible, hard exoskeleton. Sugar gliders have a relatively low protein requirement, and so excessive protein may, in fact, be detrimental to overall health. Their digestive tract comprises a simple small intestine adapted to protein and sugar digestion and a large cecum for microbial fermentation of complex carbohydrates.

The variable diet needs of sugar gliders makes for difficulty in recreating in captivity. The current recommendation for diet is through one of the diets made from a balanced recipe. In addition to these diets, small amounts of invertebrate prey, vegeta-

bles, and fruit can be offered. Examples of prepared diets include fresh nectar, maple syrup, honey, and artificial nectar mixes like Nekton Lory (Nekton Products, Germany), Roudybush Lory Nectar (Roudybush Inc., Woodland, CA, USA), and Glideraid (Exotic Nutrition, Newport News, VA, USA). Acacia gum (gum Arabic) is also commercially available. **Two of the most common mistakes made by owners are feeding a glider a diet composed of just one thing or feeding it too much of its favorite thing (typically fruits or mealworms) so they don't consume the rest of their diet. Both of these instances can lead to life threatening nutritional imbalances and periodontal disease.**

PREVENTATIVE MEDICINE AND PHYSICAL EXAMINATION

Sugar gliders can be deceptive in that they are often easy to handle, but difficult to restrain, and will vocalize loudly if you try to do things that are not on their personal agenda! To aid with restraint, a towel can be used to collect a glider from its carrier, and treats can be used to see if at least a partial cooperative physical exam can be performed. For less-cooperative sugar gliders or to assist in palpation, restraint can be applied around the back of the head and behinds the ear by the handler, with the body cupped by the rest of the hand.

Sugar gliders are not tolerant of being scruffed, and sedation or brief anesthesia may need to be utilized to allow for thorough examination. Lastly, as sugar glider are nocturnal, later afternoon appointments in minimally lit rooms will be least stressful for the animal, BUT they may be more subdued during the morning or mid afternoon, facilitating easier examination.

Like all other animals, an annual examination is always recommended for exotic pets, and sugar gliders are no exception. During annual examination, a full husbandry review should be performed to identify possible deficits in diet and husbandry and risk factors for specific diseases. There are no vaccines recommended or developed for use in sugar gliders at this time. Based upon the age of the animal, its history and its physical examination findings, one should consider some bloodwork parameters, whole body radiographs and/or fecal analysis. **Fecal direct smear is important to evaluate for any *Giardia* spp, as these animals are commonly asymptomatic carriers of this potentially zoonotic organism.**

If unable to give medications or other therapies by mouth, then various injection sites can be utilized in sugar gliders. IM injections can be performed in the epaxials of the neck/upper thorax or the biceps femoris. Subcutaneous injections can be given along dorsal midline of the thorax, **but should never be given in the patagium so as to avoid discomfort or irritation of this special structure.** IV injections are very challenging in these small animals, and though sites like the cephalic or lateral saphenous vein can be used, they should only be attempted while the animal is under sedation or anesthesia.

Phlebotomy, IV/IO catheterization, and injection sites for sugar gliders[3,11,13]

Reported Phlebotomy Sites[a]	IV Catheterization Sites[a]	IO Catheterization Sites[a]	Injection Sites
Cranial vena cava[a] Jugular vein[a] Cephalic Lateral saphenous Femoral Ventral coccygeal vein Lateral tail vein Medial tibial artery	Cephalic Lateral saphenous	Proximal tibia, through tibial crest Proximal femur, through trochanteric fossa	IM: quadriceps, gluteal, biceps, triceps SQ: intrascapular, flank IV: cephalic, lateral tail, pouch vein

COMMON DISEASES

And just as important as knowing health sugar glider information is to know about unhealthy sugar glider information in terms of common disease processes.

Self-Mutilation

If proper husbandry or socialization practices are

not followed, sugar gliders can unfortunately be quite prone to self-mutilation. Common sites include the tail, limbs, scrotum, and penis.

These behaviors may occur secondary to stress, pain/injury, internal parasites, improper nutrition, and other diseases, and can lead to serious blood loss or infection in severe cases. Examples of stress include isolation, boredom, loneliness, death of/separation from bonded cage mate(s), overcrowding, unnatural social structure, sexual frustration, unsanitary conditions, inappropriate housing or husbandry, abuse, neglect, or perceived threats (other household pets, etc.). **Gliders are also prone to self-mutilate at superficial sites of injury, such as an avulsed toenail or a constricting fur ring.** Gliders that experience self-mutilation should be assessed by a veterinarian in case treatments items such as antibiotics or analgesics are indicated. For animals that present with self-mutilation wounds, the most important first step is that of stabilization. Shocky animals should be treated with active warming, warm fluid support with dextrose if needed, and hemostasis of any actively bleeding sites. After initial stabilization, wounds can be addressed with cleaning and flushing and provision of any indicated analgesic and antibiotic support. In some cases of distal tail or reproductive trauma, amputation of the traumatized penis or tail is the best option. Castration may be helpful in reducing sexual behavior in intact males as well. A modified e-collar can TEMPORARILY be placed to help protect healing wounds, but should never be implemented as the sole treatment item for a self-mutilating sugar glider.

Nutritional Secondary Hyperparathyroidism

Gliders with nutritional secondary hyperparathyroidism can present in a couple of ways. One may be with acute hypocalcemia, which can present in the form of seizures or diffuse weakness. Other differentials for the seizing sugar glider are hypoglycemia, trauma, and encephalitis. **More chronically affected animals may present with fractures of limbs due to chronic malnutrition and poor bony development. A full husbandry review should be discussed with owners of affected animals, as 100% of these gliders have severe dietary deficits (either due to diets exclusively comprised of insects or insects/fruits).** Diagnostics that should be pursued include CBC, chemistry, ionized calcium, and radiographs to diagnose other concurrent problems related to malnutrition such as hypoglycemia, anemia, or hypoproteinemia.

Treatment for nutritional hyperparathyroidism is multifactorial. Because it is **mediated by a lack of calcium in the diet (or elevated phosphorus), correction of this is crucial with calcium supplementation.** In the acutely affected seizing animal, this should be performed with injectable calcium. However, oral calcium (calcium glubionate) can be instituted thereafter.

Because these animals have a whole body lack of calcium and are at risk for pathological fractures, strict cage rest and gentle handling is also indicated. This can be very difficult as the animal continues to feel better and wants to act normally and jump and glide. However, in the acute period, these animals should be kept in close quarters with minimal handling and food and water immediately next to the hiding box/pouch. **For the animal that presents with signs consistent with limb abnormalities or that has a fracture diagnosed, some bandaging and splints can be attempted. However, they can be difficult to keep in place without causing self-mutilatory behavior or bandage-associated morbidity. In the end, dietary correction is extremely important for recovery to**

occur and to ensure that future episodes do not recur. Counseling of clients on the intricate needs of sugar glider nutrition is important. **If insects are fed, they should be gut loaded for 24-48 hours prior with a high calcium feeder diet.**

Hypoglycemia

A condition that can also coincide with poor nutrition is hypoglycemia. Hypoglycemia can also been seen secondary to underlying hepatic or GI disease, or sepsis. Clinical signs of hypoglycemia can include lethargy, tremors, and seizures, similar to other species. Diagnostics that should be pursued include a full husbandry review, physical examination, bloodwork, and radiographs to identify comorbidities and husbandry deficits. **Treatment needs to occur rapidly, particularly in the face of seizing animals, and should include fluid therapy with additional dextrose as well as treatment of the initiating event.** Differentials for seizures are similar to those for nutritional secondary hyperparathyroidism.

Obesity

As with any pet animal, sugar gliders can also become obese in captivity, especially if fed diets too high in sugars or fat. Housing in an inadequately-sized space or other exercise-limiting environment can also predispose individuals to obesity.

Obesity can then further lead to additional comorbidities such as hepatic or cardiovascular disease. **Gravid animals can additional pass on issues to their young; lipid deposits (lipid keratopathy) can form in juvenile sugar glider's eyes if the mother is fed a diet too high in fat.** These lipid deposits appear as small white spots within the eyes that can alter visual ability.

Diarrhea

And lastly, diarrhea can be yet another common reason for presentation of a sugar glider to a veterinarian. There are causes for diarrhea including primary GI diseases and extra GI diseases. Infectious etiologies are particularly worth noting, and include bacterial (E. coli, Clostridium spp) and parasitic (nematodes, cestodes, protozoa like Giardia) organisms.

It is important to note that many sugar gliders carry a low number of *Giardia* spp. without clinical signs, and so not unlike *Salmonella* in hedgehogs, sugar glider owners should be warned when they come in for the first time that they should limit exposure by minimizing contact with children and handwashing on a routine basis. Metabolic disorders (hepatic/renal), malnutrition, and stress can also be contributing factors of diarrhea in these animals. Long-standing diarrhea places a glider at risk for rectal or cloacal prolapse, dehydration, sepsis, and hypoglycemia, so should be promptly addressed when present.

Diagnostics that should be include fecal analysis/Gram stain/+/- culture, CBC and chemistry, and radiographs. Treatment is generally supportive or focused on the primary cause. In general, fluid therapy should be initiated prior to most diagnostics to correct dehydration and support the animal. Additional supportive care may include dextrose, active warming, antibiotic/antiparasitic treatment, and supplemental feedings.

The Ten Most Important Facts

1. Hedgehogs do not hibernate, but can enter a state of dormancy called torpor. This can occur when environmental conditions are not appropriate (too hot OR too cold) and can cause depressions in mentation, heart rate, respiratory rate, and development of ataxia.
2. The anatomical structure that allows hedgehogs to curl up is collectively called the mantle, with the major muscles of the mantle being the fronto-dorsalis and caudo-dorsalis (pull the head down and rump up), panniculus carnosus (wraps around the spine), and orbicularis ("purse string" that closes everything up).
3. Hedgehogs are highly prone to development of neoplasias, with the three most common neoplasias in pet hedgehogs being oral squamous cell carcinoma, lymphosarcoma, and mammary gland tumors.
4. In pet hedgehogs presenting for dermatitis with or without spine loss, acariasis (most commonly Caparinia tripilis) is a common underlying etiology. Other causes may include dermatophytosis, allergies, or neoplasia.
5. Approximately 28% of hedgehogs are non-clinical carriers of Salmonella. In some, this microbe may cause weight loss, decreased appetite, dehydration, lethargy, and death. This is an important zoonotic pathogen and can cause serious disease in humans.
6. Ataxia is a common clinical presentation of pet hedgehogs, with the three major causes being Wobbly Hedgehog Syndrome, intervertebral disc disease, or spinal neoplasia. There is no treatment for Wobbly Hedgehog Syndrome other than supportive care.
7. The anatomical structure that allows sugar gliders to glide is called the patagium, which is a thin membrane that extends from the wrist to the ankle. Injections should NEVER be given in the patagium.
8. Sugar gliders should never be housed alone so as to avoid consequences of isolation such as self-mutilation. Like hedgehogs, sugar gliders can also enter a torpor state if housed at inappropriate temperatures.
9. Inappropriate dietary husbandry can lead to development of periodontal disease, severe nutritional deficiencies, and nutritional secondary hyperparathyroidism in sugar gliders. The latter can present as seizures, diffuse weakness, or pathologic fractures, among other signs of general systemic illness.
10. It is important to note that many sugar gliders carry a low number of Giardia spp. without clinical signs, and so sugar glider owners should be warned that they should limit exposure by minimizing contact with children and routine handwashing.